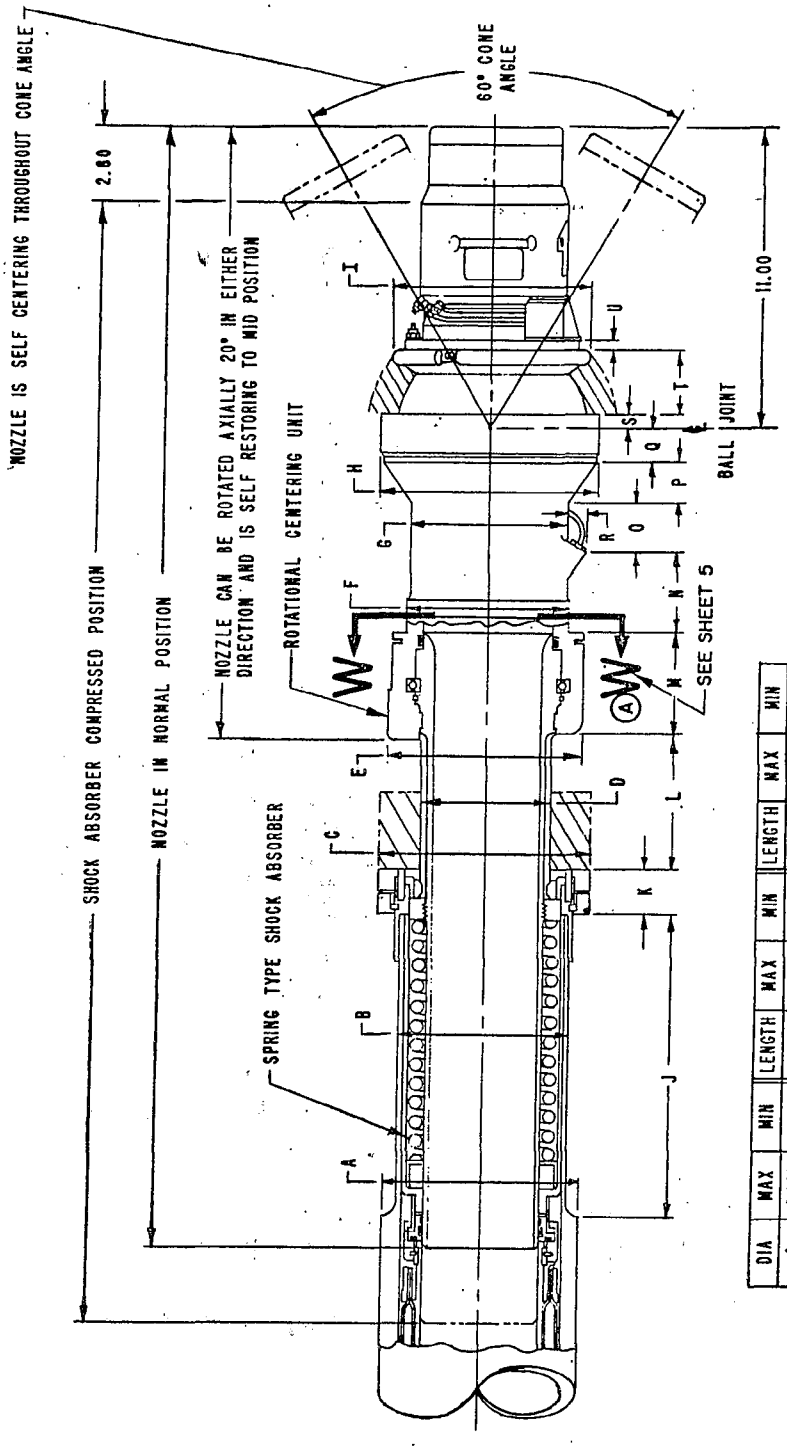
FED. SUP CLASS
1680



* DOES NOT INCLUDE SPRING GUARD OR BOLT HEADS.

ALL DIMENSIONS ARE IN INCHES.
UNLESS OTHERWISE SPECIFIED, TOLERANCE IS ± .010

DIA	MAX	MIN	LENGTH	MAX	MIN
A	6.962		J	11.175	10.970
B	6.02		K	1.680	1.555
C	7.43		L	4.959	4.629
D	4.63		M	4.070	3.744
E	6.75*		N	2.36	2.86
F	5.77		O	1.926	1.613
G	5.65		P	1.647	1.454
H	7.76		Q	1.310	1.265
I	7.05		R	.73	.67

LENGTH	MAX	MIN
S	.501	.450
T	2.463	2.386
U	.31	.30

(A) FOR CHANGES SEE SHEETS 1 AND 5.
THIS IS A DESIGN STANDARD, NOT TO BE USED AS A PART NUMBER. (A) SHEET 5 ADDED

APPROVED 11 AUGUST 69 REVISED (A) 27 Aug 71

This standard has been approved by the Department of the Air Force and is mandatory for use by that activity. All other military activities are required to employ this standard where suitable.

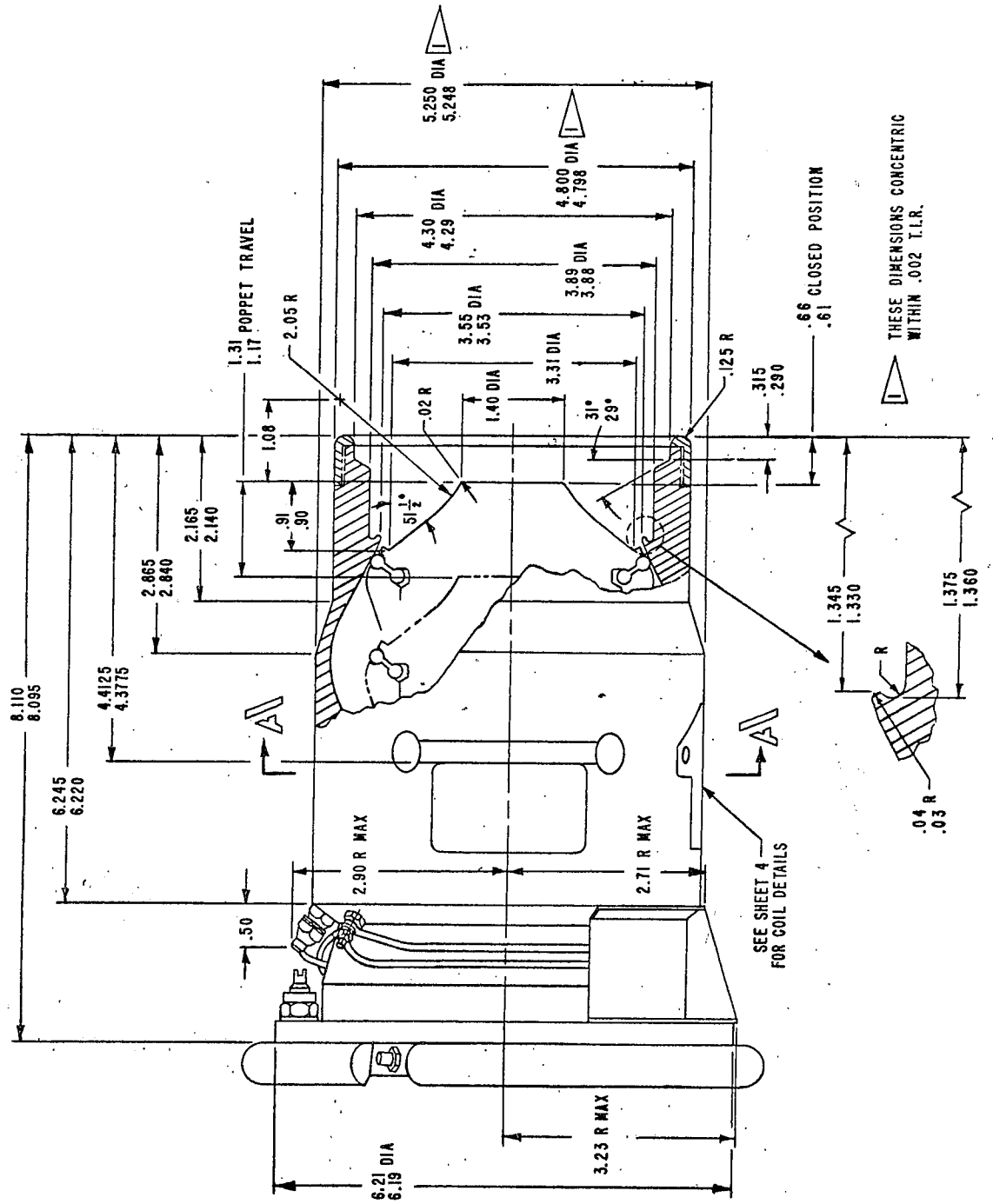
P.A. AIR FORCE 11 Other Cust	TITLE NOZZLE-UNIVERSAL AERIAL REFUELING TANKER BOOM	MILITARY STANDARD MS27604(USAF)
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 1 OF 5

DD FORM 1 SEP 69 672-1 (Limited coordination)

EDITION OF 1 SEP 67 MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:45Z
Check the source to verify that this is the current version before use.

FED. SUP CLASS
1680



APPROVED 11 AUGUST 69 REVISED (A) FOR CHANGES SEE SHEETS 1 AND 5.

This standard has been approved by the Department of the Air Force and is mandatory for use by that activity. All other military activities are required to employ this standard where suitable.

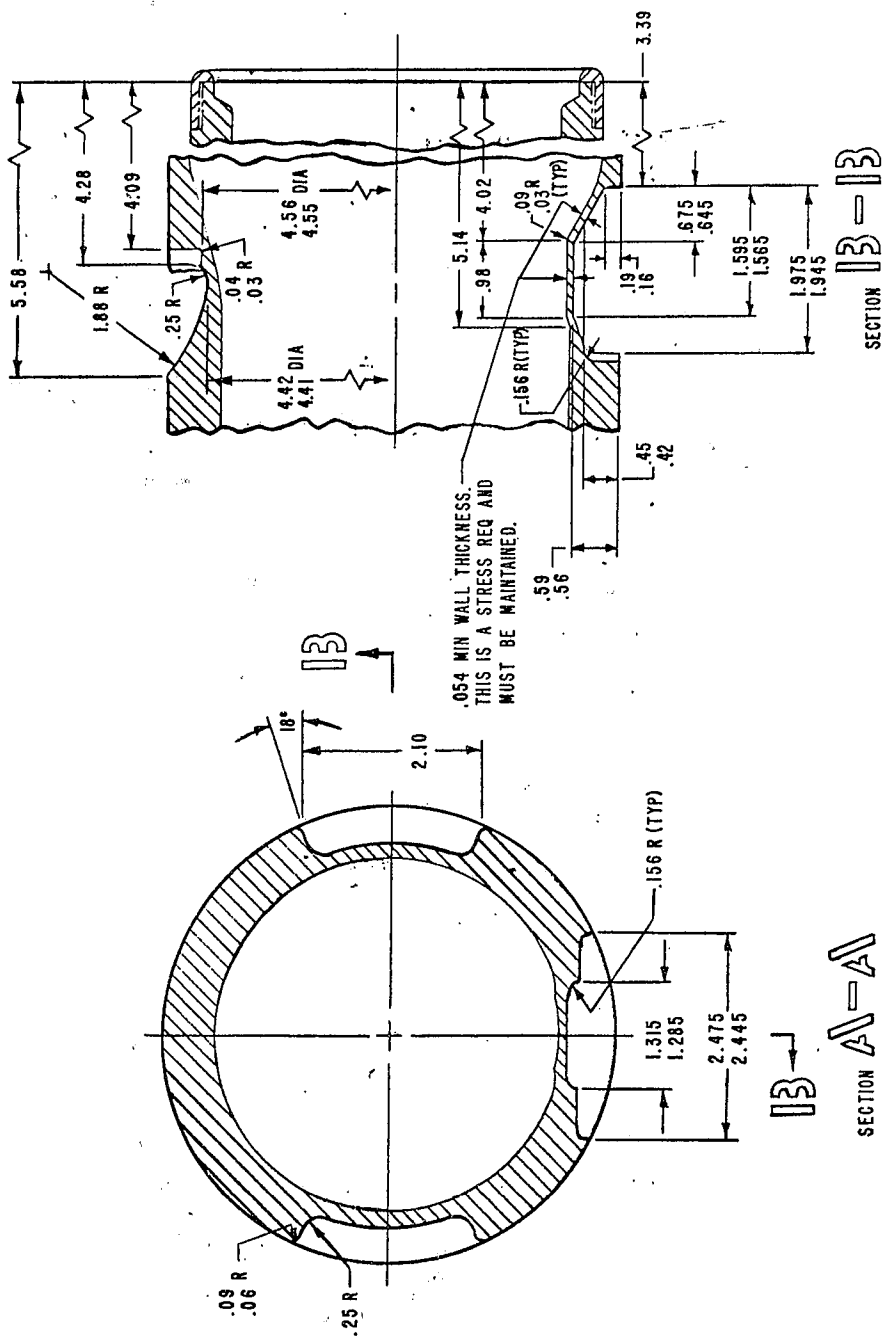
P.A. AIR FORCE 11 Other Cust	TITLE NOZZLE-UNIVERSAL AERIAL REFUELING TANKER BOOM	MILITARY STANDARD MS27604(USAF)
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 2 OF

DD FORM 1 SEP 65 672-1 (Limited coordination)

EDITION OF 1 SEP 57 MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:45Z
Check the source to verify that this is the current version before use.

FED. SUP CLASS
1680



APPROVED 11 AUGUST 69 REVISED (A) FOR CHANGES SEE SHEETS 1 AND 5.

This standard has been approved by the Department of the Air Force and is mandatory for use by that activity. All other military activities are required to employ this standard where suitable.

P.A. AIR FORCE 11 Other Cust (4)	TITLE NOZZLE-UNIVERSAL AERIAL REFUELING TANKER BOOM	MILITARY STANDARD MS27604(USAF)
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 3 OF

DD FORM 1 SEP 63 672-1 (Limited coordination)

EDITION OF 1 SEP 57 MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:45Z
Check the source to verify that this is the current version before use.

TANKER COIL REQUIREMENTS

A. COIL DESIGN

1. THE COIL SHALL BE MADE OF 2300 TURNS # 38 B AND S FORMAX MAGNETIC WIRE OR EQUIVALENT WITH A RESISTANCE OF 333 TO 432 OHMS.
2. LAMINATIONS SHALL BE 0.01 OR 0.014 THICK COMMERCIAL QUALITY TRANSFORMER STEEL WITH CORE LOSS NOT EXCEEDING 0.72 WATTS PER POUND. THE MINIMUM LAMINATION CROSS-SECTIONAL AREA SHALL BE 0.09 IN^2 WITH THE POLE FACE DIMENSIONS AS SHOWN ON FIGURE 1.

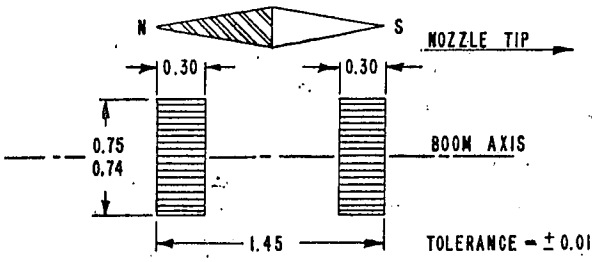


FIGURE 1

3. WHEN THE POSITIVE TERMINAL OF A 28 VOLT D.C. POWER SOURCE IS APPLIED TO THE SIGNAL INPUT LEAD THE NORTH NEEDLE OF A COMPASS SHALL POINT AS SHOWN ON FIGURE 1.

B. POLE FACE POSITIONS AND TOLERANCES

TANKER

1. THE EDGE OF THE AFT POLE FACE SHALL BE LOCATED 3.89 ± 0.10 FROM THE NOZZLE TIP AS SHOWN ON FIGURE 2.

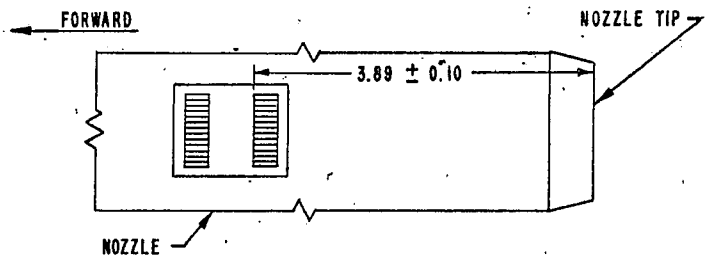


FIGURE 2

2. THE POLE FACES SHALL BE POSITIONED FROM FLUSH WITH THE NOZZLE SURFACE TO MINUS 0.01 FROM THE OUTSIDE SURFACE OF THE NOZZLE AS SHOWN ON FIGURE 3.

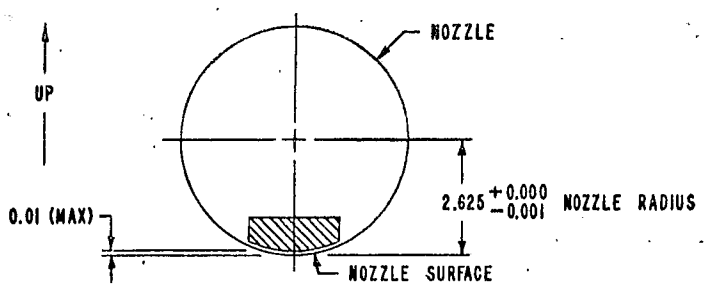


FIGURE 3

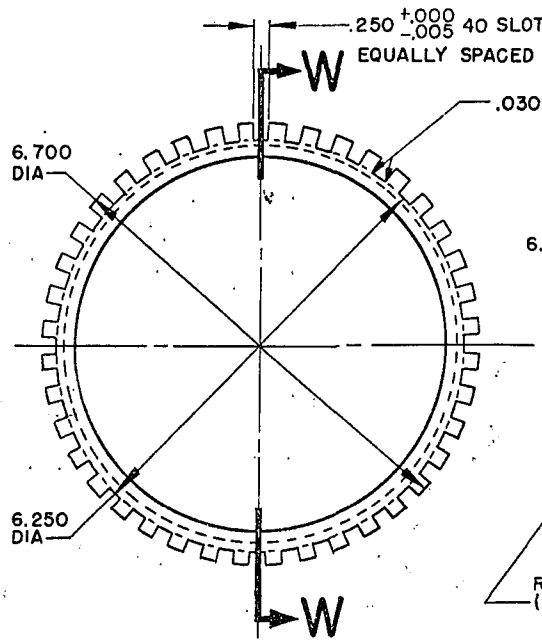
3. THE POLE FACES SHALL BE CENTERED WITHIN 0.03 TO THE LEFT OR RIGHT OF THE VERTICAL AXIS.

APPROVED 11 AUGUST 69 REVISED (A) FOR CHANGES SEE SHEETS 1 AND 2.

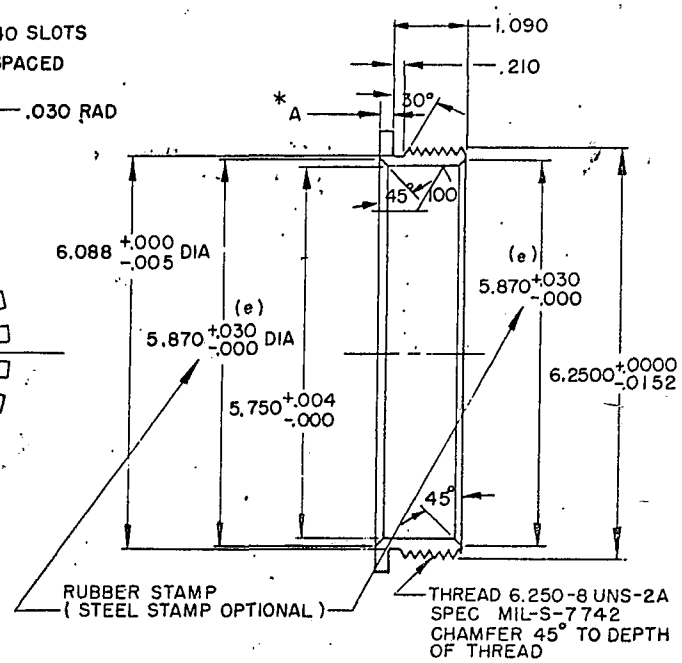
This standard has been approved by the Department of the Air Force and is mandatory for use by that activity. All other military activities are required to employ this standard where suitable.

P.A. AIR FORCE II Other Cust (a)	TITLE NOZZLE-UNIVERSAL AERIAL REFUELING TANKER BOOM	MILITARY STANDARD MS27604(USAF)
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 4 OF

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:45Z
Check the source to verify that this is the current version before use.



END VIEW



SECTION W-W

PART NO	"A" DIM
-1	.15 *
-2	.25

(a) 24S-T4 EXTRUSION PER SPECIFICATION QQ-A-200/3 CONDITION T
OPTIONAL: 24S-T4 BAR PER SPECIFICATION QQ-A-200/3 CONDITION T

(b) OPTIONAL: 14S-T6 FORGED BLOCK AMS4135 CL 3 GRAIN DIRECTION OPTIONAL

(c) 24S-F SLAB PER SPECIFICATION QQ-A-200/3 ROUGH MACHINE AND HIGH TEMPERTURE T4-FINISH MACHINE

(d) OPTIONAL: 24S-F PLATE SPECIFICATION QQ-A-250/5B AS FABRICATED. CUT TO APPROXIMATE SIZE AND HIGH TEMPERTURE TO T4 BEFORE MACHINE.

(e) LINEAR ±.010 AND OUT OF ROUNDNESS ±.003 T.I.R.

MATERIAL: STEEL, MIL-H-6875
ALUMINUM ALLOY, MIL-H-6088

FINISH: SEE PROCUREMENT SPECIFICATION

WEIGHT: APPROXIMATE .522

REMOVE BURRS, SLIVERS AND BREAK SHARP CORNERS .030 RADIUS MAXIMUM

ALL DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED

APPROVED 27 Aug 71 REVISED (A) FOR CHANGES SEE SHEETS 1 AND 5.

This standard has been approved by the Department of the that activity. All other military activities are required to employ this standard where suitable. and is mandatory for use by

P.A. USAF-11 Other Cust	TITLE NOZZLE, UNIVERSAL AERIAL REFUELING TANKER BOOM	MILITARY STANDARD MS27604(USAF)
PROCUREMENT SPECIFICATION MIL-R-27521(USAF)	SUPERSEDES:	SHEET 5 OF

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:45Z
Check the source to verify that this is the current version before use.